

LAB 4 - Linux - NTI

1. List **all running processes** and save the output to `~/processes.txt`.

```
`ps aux > ~/processes.txt`
```

2. Find the **PID (Process ID)** of the `sshd` service.

```
ps aux | grep sshd
```

3. Run a `sleep 500` command in the background, then **kill it** after 5 seconds.

```
sleep 500 &  
sleep 5 # Wait 5 seconds  
kill <PID>
```

1. Install `apache2` service, edit the default HTML file (`/var/www/html/index.html`), and verify changes in a web browser.

```
sudo apt install apache2
```

5. **Check if `sshd` (SSH service) is running**. If not, start and enable it.

```
sudo systemctl status sshd
```

6. **Restart the `cron` service** and verify its status.

```
sudo systemctl restart cron  
sudo systemctl status cron
```

7. Create a **compressed tarball** (`archive.tar.gz`) of `/var/log` and save it in your home directory.

```
sudo tar -czvf ~/archive.tar.gz /var/log
```

8. **Extract** the tarball into `~/logs\_backup/`.

```
mkdir -p ~/logs_backup  
tar -xzvf ~/archive.tar.gz -C ~/logs_backup
```

9. Create a **non-compressed tarball** (`archive.tar`) of `/etc/ssh` and save it in `/tmp`.

```
sudo tar -cvf /tmp/archive.tar /etc/ssh
```

10. Compress `~/processes.txt` using `gzip`.

```
gzip ~/processes.txt
```

11. **Decompress** it and compare file sizes using `ls -lh`.

```
gunzip ~/processes.txt.gz  
ls -lh ~/processes.txt*
```

12. **Install** `htop` (a process viewer) using your package manager.

```
sudo apt install htop
```

13. **Search** for the package `nginx` (or `httpd`) but do not install it.

```
apt search nginx
```

14. **Remove** the `vim` editor (if installed) and then reinstall it.

```
sudo apt install vim
```

15. Use `wget` to download the Linux kernel source:

```
wget https://cdn.kernel.org/pub/linux/kernel/v5.x/linux-5.19.tar.xz
```

16. Use `curl` to fetch **Google's homepage** and save it as `google.html`.

```
curl -o google.html https://www.google.com
```

17. **Create a new VM** (e.g., VirtualBox/Cloud instance), add a user to the `sudoers` group, and run `apt update && apt upgrade`.

```
udo adduser username sudo usermod -aG sudo username  
sudo apt update && sudo apt upgrade
```

18. Generate SSH key pair

```
ssh-keygen -t rsa -b 4096
```

19. Copy public key to remote server

```
ssh-copy-id username@remote_server_ip
```

20. SSH into server and verify hostname

```
ssh username@remote_server_ip hostname
```

21. Transfer archive file to remote server

```
scp ~/archive.tar.gz username@remote_server_ip:~
```